

Sex-specific relationships between gray matter volume and executive function in young children with and without prenatal alcohol exposure

Madison Long^{a,b,c}, Preeti Kar^{a,b,c}, Nils D. Forkert^{a,b,c}, Bennett A. Landman^{d,e},
Gerald F. Giesbrecht^{b,f}, Deborah Dewey^{b,c,f,h,j}, W. Ben Gibbard^{b,f}, Christina Tortorelliⁱ,
Carly A. McMorris^{b,c,f,g,h}, Yuankai Huo^{d,e}, Catherine Lebel^{a,b,c,*} 

^a Department of Radiology, University of Calgary, Calgary, Alberta, Canada

^b Owerko Centre, Alberta Children Hospital Research Institute, University of Calgary, Calgary, Alberta, Canada

^c Hotchkiss Brain Institute, University of Calgary, Calgary, Alberta, Canada

^d Department of Electrical and Computer Engineering, Vanderbilt University, USA

^e Department of Computer Science, Vanderbilt University, USA

^f Department of Paediatrics, University of Calgary, Calgary, Alberta, Canada

^g Werklund School of Education, University of Calgary, Calgary, Alberta, Canada

^h Mathison Centre for Mental Health Research and Education, Canada

ⁱ Department of Child Studies and Social Work, Mount Royal University, Calgary, Alberta, Canada

^j Department of Community Health Sciences, University of Calgary, Calgary, Alberta, Canada

ARTICLE INFO

Keywords:

Structural MRI
Executive function
Longitudinal
Prenatal alcohol exposure
Gray matter volume

ABSTRACT

Sex differences in brain volume are well established across ages however, limited research has investigated if sex differences in brain structure associate with early cognitive outcomes. Moreover, associations among sex, brain structure, and cognition in individuals with prenatal alcohol exposure (PAE), the most common known cause of developmental delay in North America, are unclear. Here, we investigated associations between executive function (measured by the BRIEF or BRIEF-P Global Executive Composite (GEC) and the Statue subtest of the NEPSY-II) and volumes of 36 gray matter regions in a longitudinal MRI sample of 169 young children ($N_{PAE}=37$; 534 total scans) aged 2–8 years. We found significant three-way interactions between sex, alcohol exposure, and executive function in 22 regions for GEC and 6 regions for Statue. Unexposed males showed negative executive function-volume associations, whereas males with PAE showed opposite associations. Unexposed females showed strong positive executive function-volume associations whereas females with PAE showed weak positive associations. We also evaluated reduced models in regions without significant 3-way interactions and found significant two-way interactions of sex and executive function for the GEC in three regions, and for the Statue in 22 regions. Males showed a negative executive function-volume relationship whereas females showed a positive relationship, regardless of exposure status. Our results suggest that males with PAE and unexposed females show relatively more mature volume-executive function relationships than females with PAE and unexposed males. This study highlights the importance of considering sex in investigations of brain and cognition, especially in populations with PAE.

1. Introduction

Average differences in brain volumes between males and females are well documented across the lifespan (DeCasien et al., 2022), but the implications they have for executive function and behaviour are not well understood. Executive function itself does not differ by sex in adulthood (Gaillard et al., 2021a; Grissom and Reyes, 2019). However, some

evidence suggests sex differences in executive function in childhood and adolescence, with males or females performing better depending on the domain measured (Brocki and Bohlin, 2004; Wierenga et al., 2019). Sex differences in executive function may be attributable to sex differences in strategies and modes of measurement, rather than ability itself (Brocki and Bohlin, 2004; Grissom and Reyes, 2019; Wierenga et al., 2014). This is further bolstered by functional MRI research showing

* Corresponding author at: Department of Radiology, University of Calgary, Calgary, Alberta, Canada
E-mail address: clebel@ucalgary.ca (C. Lebel).

<https://doi.org/10.1016/j.dcn.2025.101608>

Received 10 October 2024; Received in revised form 13 July 2025; Accepted 21 August 2025

Available online 23 August 2025

1878-9293/© 2025 Published by Elsevier Ltd. This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

task-specific sex differences in regional BOLD responses during executive function tasks such as the Iowa Gambling Task, Tower of London, Eriksen Flanker task, Go/No-Go task, and Stop Signal task, even when task performance did not differ between sexes (Gaillard et al., 2021b). However, sex differences in associations between gray matter structure and executive function in early childhood have not been characterized.

Sex differences in gray matter volume development trajectories may be limited to specific developmental periods. Early longitudinal studies examining brain development from childhood through adolescence indicated that rates of gray matter volume development differ between males and females (Lenroot et al., 2007). However, more recent evidence suggests that sex differences in development trajectories do not appear before age 8 years (Long, Kar, et al., 2024; Long, Ostertag, et al., 2024). However, the lack of sex differences in trajectories does not discount that sex differences may be present in associations between brain structure and cognition.

Executive function is a set of cognitive skills, which are strongly associated with daily functioning (Barkley, 2010; Calkins and Marcovitch, 2010; Ware et al., 2012). While many definitions of executive function have been proposed, it is typically understood that executive function is a set of higher-order cognitive functions working together to inhibit unwanted responses, switch between tasks, maintain working memory, and solve problems (Diamond, 2013; Friedman and Miyake, 2017; Zelazo et al., 2003; Zelazo and Müller, 2010). Executive function develops rapidly in early childhood, making it a construct of particular interest in young children (Anderson, 2002). Studies correlating executive function with gray matter volume in adults have generally found that larger volume is associated with better executive function (for a review, see Yuan and Raz, 2014). A similar pattern has also been observed in school-aged children and adolescents (Barch et al., 2022; McAlonan et al., 2009; Sung et al., 2021). Regarding brain structure, evidence childhood through adolescence suggests that an association between larger gray matter volume and better executive function emerges in females before it becomes evident in males (Sung et al., 2021). However, to the best of our knowledge, sex differences in brain structure and their association with executive function have not been examined in early childhood.

Impairments in executive function are commonly observed in children with developmental disorders, such as fetal alcohol spectrum disorder (FASD), which has an estimated prevalence of 2–5 % in North America (May et al., 2018; Popova et al., 2023). FASD is caused by prenatal alcohol exposure (PAE, Popova et al., 2023). PAE is one of the leading known causes of developmental delay in North America, affecting an estimated 11 % of pregnancies (Popova et al., 2023). PAE is associated with a wide range of lifelong physical and neurobehavioral difficulties, with deficits in executive function being one of the most common problems (Khouri et al., 2015; Kodituwakku et al., 2001; Mattson et al., 2019; Mattson and Riley, 1998; Rasmussen et al., 2007, 2013; Rasmussen and Bisanz, 2009; Rockhold et al., 2021). These executive function impairments are related to later outcomes such as poor school performance, difficulty with peers, mental and physical health problems, and troubles with daily functioning, such as managing money and maintaining employment (Calkins and Marcovitch, 2010; Clark et al., 2002; Ware et al., 2012).

Children with PAE are more likely than unexposed children to have additional adverse prenatal substance exposures and/or postnatal adverse experiences (Flannigan et al., 2021). A study of 1400 individuals who presented at the Washington State Fetal Alcohol Syndrome Diagnostic and Prevention Network clinic with either confirmed alcohol exposure or sentinel facial features specific to PAE reported that 93 % of the presenting individuals had at least one additional adverse prenatal substance exposure such as tobacco or illicit drugs (Astley, 2010). This same study also reported high rates of postnatal adverse experiences for individuals with PAE. At least 70 % of the participants experienced some form of postnatal adversity including multiple out-of-home placements, physical/sexual abuse, or neglect. A more recent study of 333 children

presenting at a Canadian FASD diagnostic clinic similarly suggested that adverse postnatal experiences are common for those with PAE; approximately 82 % of participants had two or more postnatal adverse experiences as measures by the Adverse Childhood Experience Questionnaire (Flannigan et al., 2021). As a result, isolating the effects of PAE on the brain is difficult. However, evidence suggests that among coexisting exposures, PAE accounts for the greatest amount of variance in structural brain differences (Hemingway et al., 2020).

PAE is associated with widespread structural brain differences, including smaller total brain, white matter, and gray matter volume (see Lebel et al., 2011 and Donald et al., 2015 for reviews). Importantly, altered brain structure has been reported for individuals with PAE in the frontal cortex (Astley et al., 2009; Rajaprakash et al., 2014), thalamus (Mattson et al., 1992; Nardelli et al., 2011; Treit et al., 2013), caudate (Archibald et al., 2001; Astley et al., 2009; Cortese et al., 2006; Mattson et al., 1996; Nardelli et al., 2011), pallidum and putamen (Nardelli et al., 2011; Roussotte et al., 2011; Treit et al., 2013), and the basal ganglia more broadly (Mattson et al., 1996). All of these regions are associated with executive function in children, adults, or both (Clark et al., 2021; Cristofori et al., 2019; Fiske and Holmboe, 2019; Friedman and Robbins, 2022; Menon and D'Esposito, 2022). A small number of studies in children with PAE have linked executive function to brain structure, including cortical thickness (Gautam, Warner, et al., 2015), surface area (Migliorini et al., 2015), white matter volume (Gautam et al., 2014), and gray matter volume (Gautam, Lebel, et al., 2015). Specifically, larger gray matter volumes were associated with poorer executive function as measured by the BRIEF Global Executive Composite in children with PAE, opposite to what was observed in unexposed children (Gautam, Lebel, et al., 2015). Additionally, a handful of studies have noted differences in functional brain activation between children with and without PAE during executive control tasks including a continuous recognition memory task and a Go/No-Go task (Burden et al., 2009, 2011; Kodali et al., 2017) and the Stop Signal task (Ware et al., 2015), suggesting that associations between brain activation and certain executive functions likely differ by PAE status. However, relationships between brain structure and executive function in very young children with PAE remain poorly understood. Moreover, prior studies did not specifically investigate sex-specific relationships, which may have implications for the relationship between brain structure and executive function, regardless of exposure status.

Across human and animal literature, differential effects of PAE on the nervous system have been noted for males and females (see reviews by Terasaki et al., 2016 and Weinberg et al., 2008). Such sex differences have also borne out in MRI studies of gray matter volume. In studies through adolescence, males with PAE display more severe subcortical gray matter volume reductions in the thalamus, putamen, and caudate when compared to unexposed males than did females with PAE when compared to unexposed females (Treit et al., 2017). In 2–3 year old children, researchers reported a similar pattern, specifically in the putamen where volume for males with PAE was greatly reduced compared to unexposed males, whereas females with PAE had similar volumes compared to unexposed females (Subramoney et al., 2022). In contrast, a recent longitudinal study using a sample overlapping with the present study (children with PAE aged 2.9–8 y) showed that females with PAE had more pronounced total and regional gray matter volume reductions than males with PAE (Long et al., 2024). In contrast to the three above-mentioned studies, Nardelli et al. found larger sex differences in gray matter volume for unexposed children than children with PAE (2015).

The present study aims to characterize the associations between regional gray matter volume and executive function in very young children (aged 2–8 years) while considering sex, PAE status, and age. We hypothesized that alcohol exposure and poorer executive function would be associated with smaller gray matter volumes in frontal and subcortical regions. Additionally, we expected to see sex differences in the strength of the relationship between executive function and gray

matter volume for the unexposed group, with females having a moderately stronger positive relationship than males. We expected more pronounced sex differences for children with PAE, with exposed females showing a positive association between gray matter volume and executive function similar to unexposed females, and exposed males showing a negative association.

2. Materials and methods

2.1. Participants

Children with PAE were recruited through caregiver support groups, early intervention services, and Alberta Children's Services in Alberta, Canada. Inclusion criteria were spoken English as a primary language, no history of head trauma, no diagnosis of autism, cerebral palsy, epilepsy or any other medical or genetic disorder associated with a serious motor or cognitive disability, and no contraindications to MRI (e.g., metal implants, dental braces). Children born at 34 weeks' gestational age or older, and those with neurodevelopmental disorders such as attention deficit hyperactivity disorder (ADHD), learning disabilities, language delays, and/or mental health diagnoses were included, as these conditions frequently co-occur with PAE (Broccia et al., 2023; Flannigan et al., 2023). Initial recruitment consisted of 57 children with confirmed PAE between ages 2 and 7 years. None of these participants were diagnosed with FASD at the time of recruitment, as diagnosis is typically not made until after 7 years of age in Alberta, Canada. PAE was confirmed via child welfare files (which included information reported from birth families, social workers, police records, and medical files), and/or using semi-structured interviews with current caregivers, case-workers, and/or birth families. 31 % of the exposed participants had confirmed PAE greater than or equal to the threshold defined in the Canadian Diagnostic Guidelines for FASD (Cook et al., 2016): ≥ 7 drinks in one week and/or two or more binge episodes (≥ 4 drinks at one time) during pregnancy. The remaining 69 % had confirmed PAE of an unspecified amount. In keeping with previous studies, 93 % of participants with PAE also had prenatal exposure to other substances. Specifically, 55 % of participants were exposed prenatally to cocaine, 45 % to cannabis, 36 % to cigarettes, 19 % to methamphetamine, 14 % to opioids, 7 % to benzodiazepines, and 5 % had drug exposure of an unspecified type. Many had overlapping exposures, and only 5 % participants had no reported prenatal drug exposures other than alcohol. Among the participants with PAE, 74 % had adverse postnatal experiences such as neglect, physical/sexual/emotional abuse, witnessing violence, witnessing substance misuse, and/or multiple caregiver transitions. The remaining 26 % of participants with PAE had no known postnatal adverse exposures. While the high degree of concurrent adverse exposures in our sample could be regarded as a confound, it also indicates that our sample is fairly representative of the larger population of individuals with PAE (Astley, 2010; Hemingway et al., 2020) and increases the ecological validity of our study. None of the participants with PAE were residing with their biological parents; all were in adoptive, foster, or kinship care. The mean age of stable placement, after which there was an absence of reported postnatal adverse experiences (as defined above), was 1.15 years and ranged from 0 to 4.1 years. 62 % children had experienced 1–2 home placements, 24 % had experienced 3–4 home placements, and 14 % experienced 5 or more placements with one child having experienced 9 different placements. For the present MRI study, children and their caregivers were invited to return approximately every 6 months for a follow-up MRI scan. Data for one child was excluded for an incidental finding on the MRI scan, and two children did not feel comfortable receiving an MRI scan. Additionally, 12 of the collected T1-weighted MRI scans were excluded due to low quality. To be included in the present study, children had at least one complete parent-report of the Behaviour Rating Inventory of Executive Function (BRIEF; Gioia et al., 2000) or BRIEF-P (Gioia et al., 2003) and/or a NEPSY-II Statue subtest score (Korkman et al., 2007), resulting

in a final analyzed dataset of 37 participants with PAE (18 male) with 75 scans (1–3 scans per subject) across the age range of 2.9–8.1 years. The mean age at scan across all sessions was 5.7 years.

The unexposed group consisted of children participating in the ongoing Alberta Pregnancy Outcomes and Nutrition (APrON) study (Kaplan et al., 2014), who had also been included in the Calgary Preschool MRI cohort (Reynolds et al., 2020, $n = 149$). Inclusion criteria for this analysis were that children were born at > 36 weeks' gestation, spoke English as a primary language, had no contraindications to MRI scans, no history of developmental delays, no presence of a neurodevelopmental disability or brain trauma, and had at least one complete parent-reported BRIEF or BRIEF-P (Gioia et al., 2000, 2003) and/or the Statue subtest of the NEPSY-II (Korkman et al., 2007). Unexposed participants had an absence of PAE and prenatal exposure to other substances confirmed via questionnaires completed by the mother during pregnancy. Additionally, unexposed participants had no reported postnatal adversities (i.e., abuse, neglect) and resided with their biological parent(s). After applying the inclusion/exclusion criteria, a total of 350 high-quality scans from 88 unexposed children (51 % male; 1–10 scans per subject) were available and used for our final analysis. Participants were invited to return for MRI scanning at 6–12 month intervals (largest inter-scan interval 36 months) across the age range of 2.6–8.5 years. For the scans included in this study, the mean age of the children at time of scan across all sessions was 5.3 years. Demographic characteristics for both exposed and unexposed groups are presented in Table 1.

2.2. Magnetic resonance image acquisition

For all children, MRI scans were completed on the same research-dedicated GE 3 T MR750w system with a 32-channel head coil at the Alberta Children's Hospital. Families were given reading and audio/visual materials to prepare children at home and were offered one or more practice sessions in an MRI simulator (Thieba et al., 2018). To minimize head motion throughout the scan, foam padding was used, and children were able to watch a movie using headphones, a projector, and a screen. The MRI protocol included the acquisition of T1-weighted anatomical images (FSPGR BRAVO sequence parameters: $0.9 \times 0.9 \times 0.9$ mm resolution, 210 axial slices, $TR = 8.23$ ms, $TE = 3.76$ ms, flip angle = 12 degrees, matrix size = 512×512 , inversion time = 540 ms). During MRI, children were not sedated and were awake viewing a movie of their choice or sleeping naturally.

2.3. Image processing

Images were initially assessed for quality at the scanner at the time of acquisition, and sequences were repeated if necessary and if time permitted. Images were also visually examined for motion after acquisition and those with major motion artifacts excluded. During processing, N4 bias corrected images (Tustison et al., 2010) were resampled to a voxel size of 1 mm^3 in preparation for multi-atlas segmentation combined with cortical reconstruction using implicit surface evolution (MaCRUISE; Huo, Carass, et al., 2016; Huo et al., 2017; Huo, Plassard, et al., 2016). MaCRUISE integrates the processes of cortical reconstruction and multi-atlas segmentation to produce reliable and consistent cortical surface parcellations in anatomical agreement with brain segmentations (Huo, Carass, et al., 2016; Huo et al., 2017; Huo, Plassard, et al., 2016). In the MaCRUISE pipeline, skull and dura-stripped images are subject to both multi-atlas segmentation of 132 regions (Asman and Landman, 2012, 2013; Klein et al., 2010) and TOPOLOGY-preserving Anatomical Segmentation (TOADS) fuzzy membership segmentation (Bazin and Pham, 2008). MaCRUISE then fuses the rigid multi-atlas and TOADS segmentations, resulting in a full cerebrum segmentation comprised of a gray matter and white matter component. To achieve a cortical reconstruction consistent with the segmentations, MaCRUISE applies multi-atlas anatomically consistent gray matter enhancement (MaACE; Han and Fischl, 2007; Sethian,

Table 1

Description of the present sample by PAE status and sex. For the PAE group, maternal education, marital status, and family income refer to the child's present caregivers/family.

		PAE		Unexposed	PAE v. Unexposed (χ^2)
	No. Participants	37		88	
	Sex	Male (49 %)	Female (51 %)	Male (51 %)	Female (49 %)
	ADHD Diagnosis	6 (16 %)		0	
		3	3	0	0
	T-Score above BRIEF/BRIEF-P Clinical Cutoff	26 (70 %)		6 (7 %)	
		14	12	5	1
Maternal Education	High School Diploma	4 (11 %)		1 (1 %)	< 0.001
		3	1	1	
	Some post-secondary	5 (14 %)		5 (6 %)	
		3	2	1	
	Trade/Technical Diploma	5 (14 %)		11 (13 %)	
		2	3	3	
	Bachelor's Degree	5 (14 %)		43 (49 %)	
		3	2	23	
	Some post-graduate or Graduate degree	3 (8 %)		28 (32 %)	
		2	1	17	11
	Missing	15		0	
Marital Status	Single/Never Married	2 (5 %)		1 (1 %)	< 0.01
		1	1	0	
	Married/Common Law	21 (57 %)		86 (98 %)	
		11	10	44	
	Divorced/Separated	5 (14 %)		1 (1 %)	
		2	3	1	0
	Missing	9		0	
Family Income	Up to 49,999	8 (22 %)		0	< 0.001
		3	5	0	
	50,000–74,999	11 (30 %)		2 (2 %)	
		4	7	1	
	75,000–99,999	6 (16 %)		19 (22 %)	
		6	0	8	
	100,000–124,999	9 (24 %)		13 (15 %)	
		4	5	6	
	125,000–149,999	0		4 (5 %)	
		0	0	1	
	150,000–174,999	1 (3 %)		17 (19 %)	
		0	1	10	
	175,000	0		32 (36 %)	
		0	0	18	
	Missing	2		1	

1999) to the gray matter component while applying a topology correction to the white matter component (Han et al., 2001, 2002).

These refined gray and white matter segmentations form the outer and inner surfaces of the reconstructed cortex, respectively. Lastly, to resolve any remaining disagreements between the multi-atlas segmentation and reconstructed surfaces, MaCRUISE refines boundaries in the multi-atlas segmentation using the inner and outer cortical surfaces (Huo, Plassard, et al., 2016). We extracted the refined segmentations (in ml) for analyses of regional volume. After automatic segmentation, trained raters checked the segmentations for accuracy and assigned them a quality score of 1 (poor), 2 (unsatisfactory), 3 (satisfactory), and 4 (excellent). Segmentations with a quality score < 3 were manually edited and reintroduced to the MaCRUISE pipeline at the segmentation fusion step, in place of the original rigid multi-atlas segmentation. Edited segmentation outputs were reassessed for quality and included in the analysis if the resulting segmentation obtained a quality score of 3 or 4. At each of the quality assessment stages, raters were blinded to the group membership of individual participants to reduce bias. In total, 87 % of the unexposed group scans and 81 % of scans from the PAE group were retained for analysis (final sample sizes: unexposed=88, PAE=37). Lastly, we performed a longitudinal registration of the data to achieve an improved biological plausibility of volume changes between timepoints and prevent additive effects of small segmentation errors on the longitudinal analysis. Therefore, the T1-weighted image from the most recent timepoint for each subject was registered to each previous timepoint employing a non-linear transformation using NiftyReg software (NiftyReg, 2023). Next, the resulting non-linear transformation fields

were applied to the segmentation image for the most recent timepoint, warping the final segmentation to the image space of each previous timepoint using a nearest-neighbour interpolation. The choice to register the oldest-age image for a subject to each previous timepoint is based in the principle that older children generally move less during scan acquisition and thus provide higher quality images than scans from a younger child. Using the oldest available timepoint as each subject's floating image increases the likelihood that subsequent registrations are based on the highest quality data available for each subject while maintaining a consistent registration procedure across all subjects. Final volume measurements were computed in the original T1 image space for each timepoint. For this study, we selected 18 regions of interest in each hemisphere (36 total regions) a-priori based on a literature review of gray matter regions involved in executive function. These regions were the right and left accumbens, caudate, pallidum, putamen, thalamus, anterior cingulate, middle cingulate, posterior cingulate, anterior insula, posterior insula, superior frontal gyrus, medial superior frontal gyrus, anterior orbital gyrus, frontal operculum, frontal pole, medial frontal cortex, middle frontal gyrus, and inferior frontal gyrus. Spaghetti plots of the final volume values were visually inspected to ensure plausibility of inter-scan volume changes and identify outliers. Doing so, we identified two potential outliers for gray matter volume; one unexposed participant had an unexpectedly large thalamus volume and one participant with PAE had an unusually large left middle frontal gyrus and superior frontal gyrus. We visually inspected T1-weighted images and associated segmentations from these participants and determined that the images were of sufficient quality, fair representations of the

subjects' anatomy, and could therefore remain included in the analysis. To further bolster our confidence in including the outliers, analyses for the affected regions were run with the respective outliers excluded. Inclusion or exclusion of the outliers did not affect the outcome of model selection (see model selection procedure in Section 2.5) and thus were included in the final analyses presented in this manuscript.

2.4. Measures of executive function

Executive function was measured using parent report on the Behavior Rating Inventory of Executive Function (BRIEF; Gioia et al., 2000) or BRIEF-Preschool Version (BRIEF-P; Gioia et al., 2003; Sherman and Brooks, 2010). These questionnaires assess executive functions in daily life situations and have been used to parse executive function ability in both clinical and non-clinical populations of children (Gioia et al., 2000, 2002, 2003). Parents of children in the unexposed sample completed the BRIEF-P or BRIEF for their child one or two times between ages 3–6 years as part of the AprON study (Kaplan et al., 2014). Caregivers of children in the PAE sample completed either the BRIEF-P or BRIEF at time of MRI scanning. The BRIEF and BRIEF-P include age- and sex-standardized subscales that capture different domains of executive function (e.g., working memory, inhibitory control, cognitive flexibility). The use of age-standardized *T* scores helps mitigate variation introduced by the difference in measurement acquisition timing for our PAE and unexposed cohorts. Global Executive Composite (GEC) is the summary score that incorporates all of the subscales to provide an overall estimate of executive function skills. Given that the BRIEF and BRIEF-P subscales differ by age (i.e., they are age appropriate), this study used only the GEC *T* scores. Raw scores are converted to age and sex-normed *T* scores, which have a standardized mean of 50 with a standard deviation of 10. Lower *T* scores on the GEC indicate fewer executive function problems. Scales of the BRIEF-P have a clinical cut-off *T* score of 65, with scores above 65 indicating clinically significant executive functioning difficulties. The clinical cut-off for the BRIEF is 70. To maximize the number of participants included in the study, children were included if they had at least one complete BRIEF-P or BRIEF questionnaire. Although some participants had multiple BRIEF or BRIEF-P measurements, we chose to include only measurement per subject to reduce model complexity and maintain greater statistical power. For participants with multiple measurements, we examined whether there were significant changes in BRIEF or BRIEF-P GEC scores over time and found no significant effect of time in either the PAE or unexposed groups. A single BRIEF-P or BRIEF measurement was then randomly selected from each participant for use in the analysis.

Executive function was also assessed with the Statue subtest of the NEPSY-II (Korkman et al., 2007), which measures children's motor persistence and motor response inhibition in a structured setting and is an age-appropriate behavioural measure of executive function for this sample. This subtest requires the child to stand motionless for 75 s with their eyes closed, ignoring distractions created by the examiner. The examiner observes and records any movements and any eye opening. Raw scores are standardized by child age resulting in a mean scaled score of 10 and standard deviation of 3. Higher scores indicate better performance, whereas lower scores indicate more task errors (e.g., body movement, eyes opened). Children in the unexposed sample completed the Statue test one or two times between ages 3–6 years as part of the AprON study (Kaplan et al., 2014). Children in the PAE sample completed the Statue test at time of MRI scanning. Participants were included if they had at least one complete Statue test. Similar to the BRIEF/BRIEF-P, we examined whether there were significant changes in Statue scores over time for subjects with multiple measurements and found no significant effect of time in either the PAE or unexposed groups. A single Statue measurement was then randomly selected from each participant for use in the analysis.

The GEC is a parent report of executive function, and the Statue subtest is a child performance measure of executive function. Previous

research suggest that parent reports and child performance measures assess different constructs of executive function (Ten Eyck and Dewey, 2016). Used together, they may provide a more complete picture of children's executive function skills than one measure alone.

2.5. Statistical analysis

All data visualizations and analyses were performed in R (R Core Team, 2023). Mixed effects models with fixed slopes and random intercepts by subject were constructed using the lme4 package (Bates et al., 2015). All variables were z-score transformed to enable extraction of standardized beta values and post-hoc comparison of interaction effects with the emmeans package (Lenth, 2017) package. We first conducted Mann Whitney U-tests to examine group differences in Statue and GEC scores. Next, for each of the executive function measures (GEC and Statue), preliminary data visualization and analysis showed that age-by-executive function score and age²-by-executive function score interactions were weak with small effect sizes ($\eta_p^2 < 0.06$). This confirmed that brain development trajectories did not vary by the age-standardized GEC or Statue scores in this sample. To determine a set of final models to use to investigate the associations between PAE, executive function, sex, and gray matter volume, we began with the following quadratic, 3-way interaction model:

Model 1

$$Y_{ij} = (B_1 \cdot a_{ij}) + (B_2 \cdot a_{ij}^2) + (B_6 \cdot P_{ij} \cdot x_{ij}) + (B_7 \cdot S_{ij} \cdot x_{ij}) + (B_8 \cdot P_{ij} \cdot S_{ij}) \\ + (B_9 \cdot P_{ij} \cdot S_{ij} \cdot x_{ij}) + (B_3 \cdot P_{ij} + B_4 \cdot S_{ij} + B_5 \cdot x_{ij} + B_{0i}) + \epsilon_{ij}$$

Where for each region of interest, Y_{ij} = the volume measurement for the j^{th} timepoint for the i^{th} subject, a_{ij} = subject's age at time of scan, x_{ij} = subject's executive function score, P_{ij} = group (PAE=1, unexposed=0), S_{ij} = subject's sex (female=1, male=0), B_{0i} = subject-specific coefficient for the y-intercept, B_1 = coefficient for age, B_2 = coefficient for age², B_3 = coefficient for group main effect, B_4 = coefficient for sex main effect, B_5 = coefficient for executive function score main effect, B_6 = coefficient for PAE by executive function score interaction, B_7 = coefficient for sex by executive function interaction, B_8 = coefficient for PAE by sex interaction, B_9 = coefficient for PAE by sex by executive function interaction, and ϵ_{ij} = random error. Consistent notation is used in each of the subsequent models.

If the 3-way interaction in model 1 had a p-value > 0.05, it was removed from the model along with the PAE by sex interaction term, which is included in Model 1 by necessity but is not of interest to the present study as our group previously characterized associations of PAE and child sex with gray matter volume in an overlapping sample (See Long, Kar, et al., 2024). We then evaluated the following reduced model:

Model 2

$$Y_{ij} = (B_1 \cdot a_{ij}) + (B_2 \cdot a_{ij}^2) + (B_6 \cdot P_{ij} \cdot x_{ij}) + (B_7 \cdot S_{ij} \cdot x_{ij}) + (B_3 \cdot P_{ij} \\ + B_4 \cdot S_{ij} + B_5 \cdot x_{ij} + B_{0i}) + \epsilon_{ij}$$

No region had both a significant interaction of sex with executive function, and PAE status with executive function so, a third set of models was fit. For each region, the 2-way interaction with the highest p-value was removed and one of the models below was computed for each region.

Model 3a (PAE by executive function retained)

$$Y_{ij} = (B_1 \cdot a_{ij}) + (B_2 \cdot a_{ij}^2) + (B_6 \cdot P_{ij} \cdot x_{ij}) + (B_3 \cdot P_{ij} + B_4 \cdot S_{ij} + B_5 \cdot x_{ij} + B_{0i}) + \epsilon_{ij}$$

Model 3b (Sex by executive function retained)

$$Y_{ij} = (B_1 \cdot a_{ij}) + (B_2 \cdot a_{ij}^2) + (B_7 \cdot S_{ij} \cdot x_{ij}) + (B_3 \cdot P_{ij} + B_4 \cdot S_{ij} + B_5 \cdot x_{ij} + B_{0i}) \\ + \epsilon_{ij}$$

In regions where models 3a or 3b had no 2-way interaction with p-value < 0.05, we computed main effects only models and considered

this to be the final model for the region.

Model 4

$$Y_{ij} = (B_1 \cdot a_{ij}) + (B_2 \cdot a_{ij}^2) + (B_3 \cdot P_{ij} + B_4 \cdot S_{ij} + B_5 \cdot X_{ij} + B_{0i}) + e_{ij}$$

False discovery rate (FDR) correction (resulting in q-values) was applied to each group of final models using the Benjamini-Hochberg method (Benjamini and Hochberg, 1995). For analyses with the BRIEF/BRIEF-P, we corrected across 22 p-values in the Model 1 group, 3 p-values in the Model 3 group, and 11 p-values in the Model 4 group (36 total). In the analyses with the NEPSY-II Statue, we corrected across 6 p-values in the Model 1 group, 22 p-values in the Model 3 group, and 8 p-values in the Model 4 group.

3. Results

Statue scores were similar across groups (unexposed vs. PAE: $W = 2238$, $p = 0.06$) and between males and females within exposure groups (unexposed male vs. unexposed female: $W = 952.5$, $p = 0.08$; PAE male vs. PAE female: $W = 147$, $p = 0.33$; see Fig. 1). GEC scores were higher in the PAE group (unexposed vs. PAE: $W = 347$, $p < 0.001$), indicating worse executive function. GEC scores were similar between sexes within exposure groups (unexposed male vs. unexposed female: $W = 1047$, $p = 0.39$; PAE male vs. PAE female: $W = 166$, $p = 0.68$). Additionally we performed Spearman correlations between age at stable placement and each of the executive function measures to help characterize the relationship between postnatal adversity and executive function outcomes in our sample. We found weak correlations between age at stable placement and NEPSY-II Statue ($r = -0.05$, $p = 0.74$) and for the BRIEF/BRIEF-P GEC ($r = 0.17$, $p = 0.32$).

3.1. Three-way interaction (sex by executive function by PAE) final models

The three-way interaction model was the best fitting model for 22

regions in the GEC analysis and 6 regions in the Statue analysis (Tables 2, 3).

3.1.1. Sex by GEC by PAE

We found significant three-way interactions (Fig. 2, Table S1) in the bilateral pallidum ($q_{left} < 0.005$, $\eta_{left}^2 = 0.06$; $q_{right} = 0.012$, $\eta_{right}^2 = 0.04$), anterior cingulate gyrus ($q_{left} < 0.005$, $\eta_{left}^2 = 0.09$; $q_{right} = 0.001$, $\eta_{right}^2 = 0.1$), middle cingulate gyrus ($q_{left} = 0.002$, $\eta_{left}^2 = 0.07$; $q_{right} = 0.013$, $\eta_{right}^2 = 0.04$), posterior cingulate gyrus ($q_{left} = 0.004$, $\eta_{left}^2 = 0.06$; $q_{right} = 0.007$, $\eta_{right}^2 = 0.06$), anterior insula ($q_{left} = 0.004$, $\eta_{left}^2 = 0.06$; $q_{right} < 0.05$, $\eta_{right}^2 = 0.03$), posterior insula ($q_{left} = 0.007$, $\eta_{left}^2 = 0.05$; $q_{right} = 0.009$, $\eta_{right}^2 = 0.05$), middle frontal gyrus ($q_{left} = 0.002$, $\eta_{left}^2 = 0.08$; $q_{right} = 0.002$, $\eta_{right}^2 = 0.08$), superior frontal gyrus ($q_{left} = 0.001$, $\eta_{left}^2 = 0.09$; $q_{right} = 0.003$, $\eta_{right}^2 = 0.06$), medial superior frontal gyrus ($q_{left} = 0.007$, $\eta_{left}^2 = 0.05$; $q_{right} = 0.007$, $\eta_{right}^2 = 0.06$), and inferior frontal gyrus ($q_{left} = 0.03$, $\eta_{left}^2 = 0.03$; $q_{right} < 0.05$, $\eta_{right}^2 = 0.03$), as well as the left anterior orbital gyrus ($q = 0.007$, $\eta_p^2 = 0.05$), and medial frontal cortex ($q = 0.02$, $\eta_p^2 = 0.03$). For all regions, unexposed females showed a negative relationship between volume and GEC T core (mean $\beta = -0.42$), such that smaller volumes were associated with more executive function difficulties. Females with PAE showed weak negative (17/22 regions, mean $\beta = -0.12$) or nearly null positive (4/22 regions, mean $\beta = 0.04$) associations between brain volumes and GEC T scores. The pattern differed slightly in the left superior frontal gyrus, where females with PAE showed a positive association between volume and GEC score ($\beta = 0.19$). In 21/22 regions, unexposed males showed positive associations: volume was positively associated with the GEC T score (mean $\beta = 0.21$). The one exception was the left inferior frontal gyrus, where unexposed males displayed a weak, negative association between volume and GEC T score ($\beta = -0.06$). Males with PAE primarily showed opposite associations compared to those observed in unexposed males, with strong negative associations between volumes and GEC T scores (mean $\beta = -0.499$).

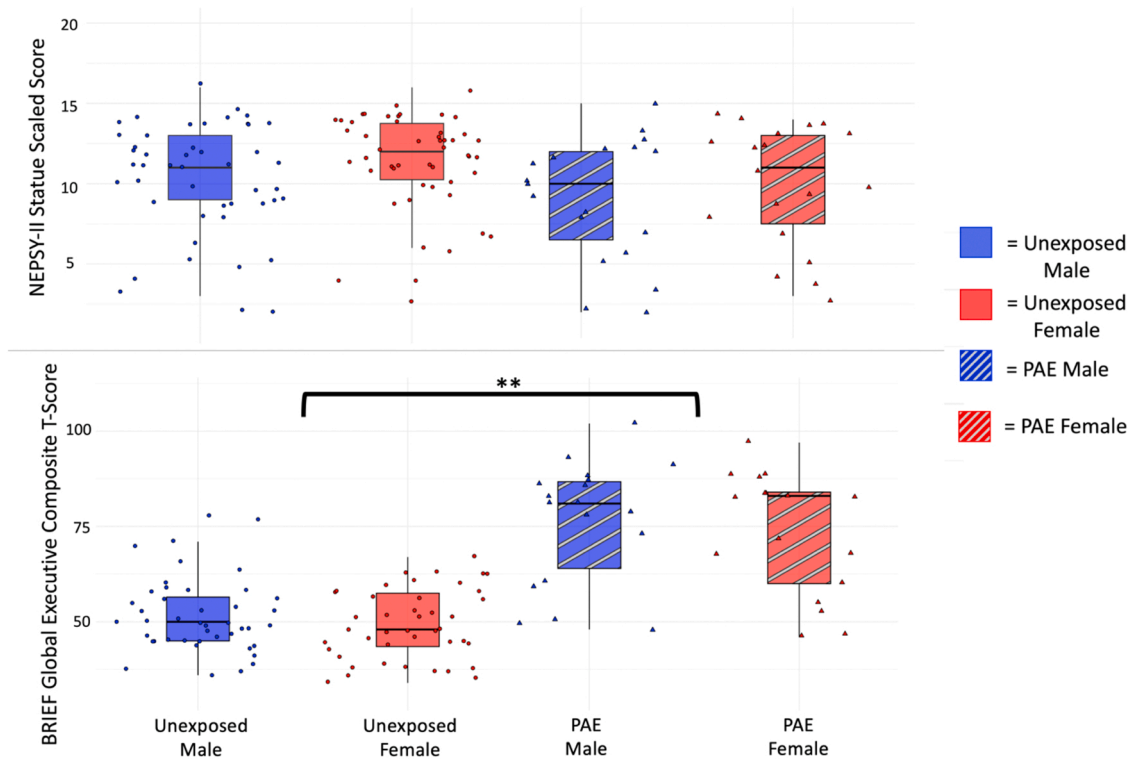


Fig. 1. Box plot showing the distribution of scores for the Statue and the GEC. We found no significant differences in Statue scores between exposure groups. For the GEC, there was a significant difference (indicated by **) between the unexposed and PAE groups. Within exposure groups there were no sex differences in performance on executive function measures.

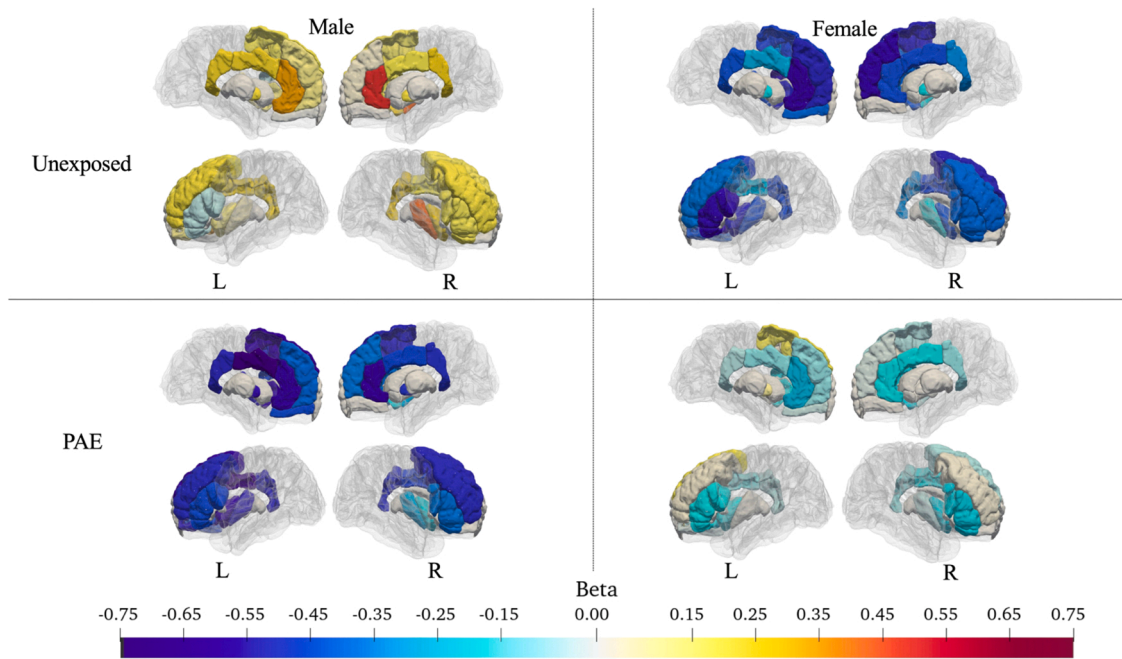


Fig. 2. Beta values in regions with significant 3-way interactions of sex, PAE, and GEC score for each group: unexposed males (top left), unexposed females (top right), males with PAE (bottom left), and females with PAE (bottom right). Regions tested in the analysis with non-significant 3-way interactions are coded as having a beta value of 0.

3.1.2. Sex by Statue by PAE

In the Statue analyses, we found significant three-way interactions (Fig. 3, Table S2) bilaterally in the anterior cingulate gyrus ($q_{left} = 0.02$, $\eta^2_{left} = 0.06$; $q_{right} < 0.05$, $\eta^2_{right} = 0.03$) and middle cingulate gyrus ($q_{left} = 0.03$, $\eta^2_{left} = 0.04$; $q_{right} < 0.05$, $\eta^2_{right} = 0.03$), as well as the right superior frontal gyrus ($q < 0.05$, $\eta^2_p = 0.03$), and left middle frontal gyrus ($q < 0.05$, $\eta^2_p = 0.03$). Unexposed females showed a positive relationship between volume and Statue scores in all six regions (mean $\beta = 0.33$), indicating that larger volumes are associated with better performance

on the Statue test. For females with PAE negative associations were found between regional volumes and Statue scores in the left anterior cingulate, bilateral middle cingulate, and right superior frontal gyrus (mean $\beta = -0.17$) and very weak positive associations were noted in the right anterior cingulate and left middle frontal gyrus (mean $\beta = 0.07$). Unexposed males showed negative associations between volumes and Statue scores in all six regions (mean $\beta = -0.26$). In males with PAE, very weak positive associations were found between volumes and Statue score in the right anterior cingulate, left middle cingulate, left middle

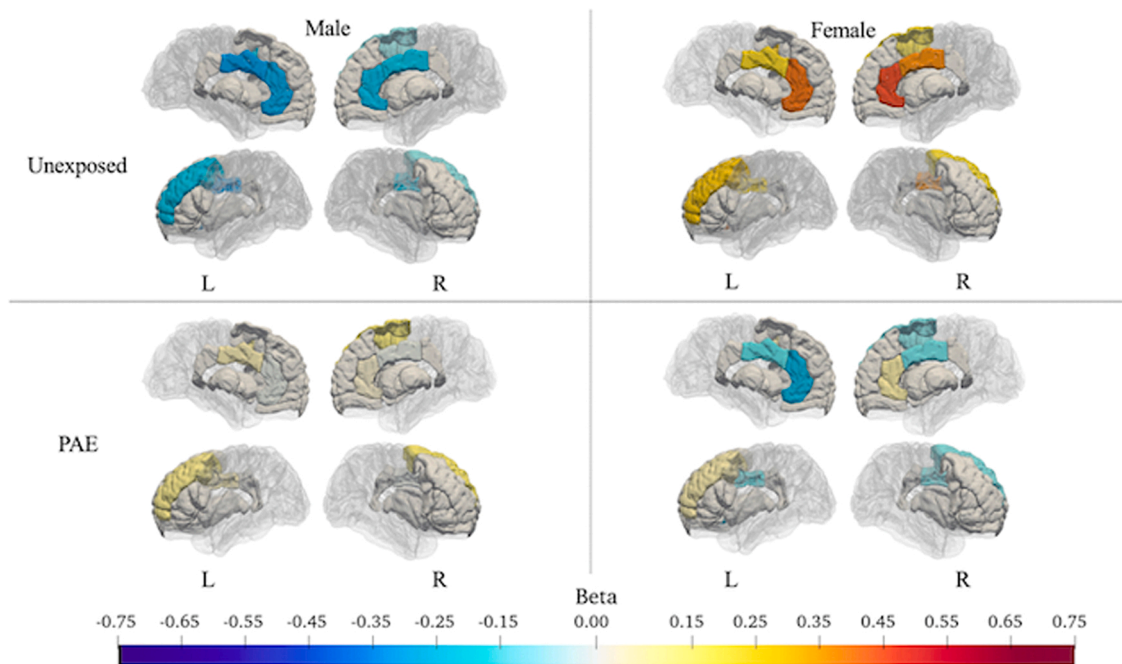


Fig. 3. Beta values in regions with significant 3-way interactions of sex, PAE, and Statue score for each group: unexposed males (top left), unexposed females (top right), males with PAE (bottom left), and females with PAE (bottom right). Regions tested in the analysis with non-significant 3-way interactions are coded as having a beta value of 0.

frontal gyrus, and right superior frontal gyrus (mean $\beta = 0.09$), and nearly null negative associations in the left anterior cingulate and right middle cingulate (mean $\beta = -0.01$)

3.2. Two-way interaction (sex by executive function) final models

Model 3b, a 2-way interaction model of sex and executive function, was selected as the final model for three regions in the GEC analyses and 22 regions in the Statue analysis.

3.2.1. Sex by GEC

We found significant sex-GEC score interactions (Table S3) in the right medial frontal cortex ($q = 0.02$, $\eta_p^2 = 0.03$), the left thalamus ($q = 0.03$, $\eta_p^2 = 0.02$), and frontal pole ($q = 0.04$, $\eta_p^2 = 0.04$). Females had moderate negative associations between volumes and GEC T scores (mean $\beta = -0.34$). For males, the associations were also negative (mean $\beta = -0.09$).

3.2.2. Sex by Statue

In the Statue analysis, we found significant two-way interactions (Fig. 4, Table S4) bilaterally in the pallidum ($q_{\text{left}} = 0.004$, $\eta_{\text{left}}^2 = 0.06$; $q_{\text{right}} = 0.012$, $\eta_{\text{right}}^2 = 0.04$), putamen ($q_{\text{left}} = 0.02$, $\eta_{\text{left}}^2 = 0.06$; $q_{\text{right}} < 0.05$, $\eta_{\text{right}}^2 = 0.03$), inferior frontal gyrus ($q_{\text{left}} = 0.04$, $\eta_{\text{left}}^2 = 0.02$; $q_{\text{right}} = 0.002$, $\eta_{\text{right}}^2 = 0.06$), anterior insula ($q_{\text{left}} = 0.016$, $\eta_{\text{left}}^2 = 0.03$; $q_{\text{right}} = 0.002$, $\eta_{\text{right}}^2 = 0.06$), posterior insula ($q_{\text{left}} = 0.02$, $\eta_{\text{left}}^2 = 0.03$; $q_{\text{right}} < 0.001$, $\eta_{\text{right}}^2 = 0.1$), medial frontal cortex ($q_{\text{left}} < 0.001$, $\eta_{\text{left}}^2 = 0.07$; $q_{\text{right}} = 0.04$, $\eta_{\text{right}}^2 = 0.02$), medial superior frontal gyrus ($q_{\text{left}} = 0.006$, $\eta_{\text{left}}^2 = 0.04$; $q_{\text{right}} = 0.004$, $\eta_{\text{right}}^2 = 0.05$), and posterior cingulate gyrus ($q_{\text{left}} < 0.001$, $\eta_{\text{left}}^2 = 0.09$; $q_{\text{right}} < 0.001$, $\eta_{\text{right}}^2 = 0.09$), as well as in the right accumbens ($q = 0.01$, $\eta_p^2 = 0.04$), caudate ($q = 0.02$, $\eta_p^2 = 0.03$), and middle frontal gyrus ($q < 0.001$, $\eta_p^2 = 0.09$), anterior orbital gyrus ($q = 0.002$, $\eta_p^2 = 0.06$), and the left frontal pole ($q = 0.004$, $\eta_p^2 = 0.05$) and superior frontal gyrus ($q = 0.03$, $\eta_p^2 = 0.03$). Across all 22 regions, females had positive associations between volumes and NEPSY-II Statue scores (mean $\beta = 0.23$), while males showed a negative or nearly null associations in most regions (mean $\beta = -0.17$).

4. Discussion

We investigated sex-specific associations between regional gray matter volumes and executive function in young children with and without PAE. Our study has three key findings. First, female children with and without PAE showed different relationships between gray matter volumes and executive function. Unexposed females showed a

strong positive relationship between regional volumes and executive function ability, whereas females with PAE showed a more regionally variable pattern with overall weaker volume-executive function associations. Second, unexposed males primarily showed negative volume-executive function association such that smaller volumes were related to better executive function, whereas males with PAE showed positive associations. The first two findings were primarily noted when executive function was measured by parent report (GEC) and suggest that PAE alters brain-behaviour relationships differently for males and females in everyday contexts. Our third key finding was that some regions showed sex differences in the volume-executive function associations independent of PAE status (*i.e.*, 2-way interaction models), especially when executive function was measured by a child performance measure (Statue subtest).

4.1. Differences by PAE status in female children

Unexposed females consistently exhibited negative associations between volumes and GEC T scores, and a positive relationship between Statue subtest scores and volume. Higher GEC scores and lower scores on the Statue subtest are indicative of poorer executive function skills, therefore our findings suggest that larger brain volumes are associated with better executive function in unexposed females. For the GEC analysis, females with PAE displayed primarily weak negative associations between volumes and GEC scores, suggesting a similar but weaker pattern compared to that of unexposed females. For the Statue subtest, females with PAE again showed weak associations with gray matter volumes, but the directions of these associations were regionally mixed. Difference in patterns of the associations between the GEC and Statue analyses for females with PAE could be partly attributed to the difference in how the GEC and Statue are measured (*i.e.*, parent report vs. performance-based) and suggests that the nature of the volume-executive function relationship for females with PAE could be dependent on how executive function is measured. Previous research on relationships between gray matter volume and executive function, which has largely focused on adolescents and adults, generally indicates that larger gray matter volume is associated with better executive function across various tasks, including the self-reported Urgency-Premeditation-Perseverance-Sensation Seeking (UPPS) Impulsive Behaviour scale and the Delis-Kaplan executive function system card sorting test (Müller et al., 2015), anti-saccade, category switching and keep track tasks (Smolker et al., 2015), Stroop test (Sung et al., 2021; Takeuchi et al., 2017), Trail Making Test and Controlled Oral Word Association test, (Newman et al., 2007), Tests of Everyday Attention (Lean et al., 2017),

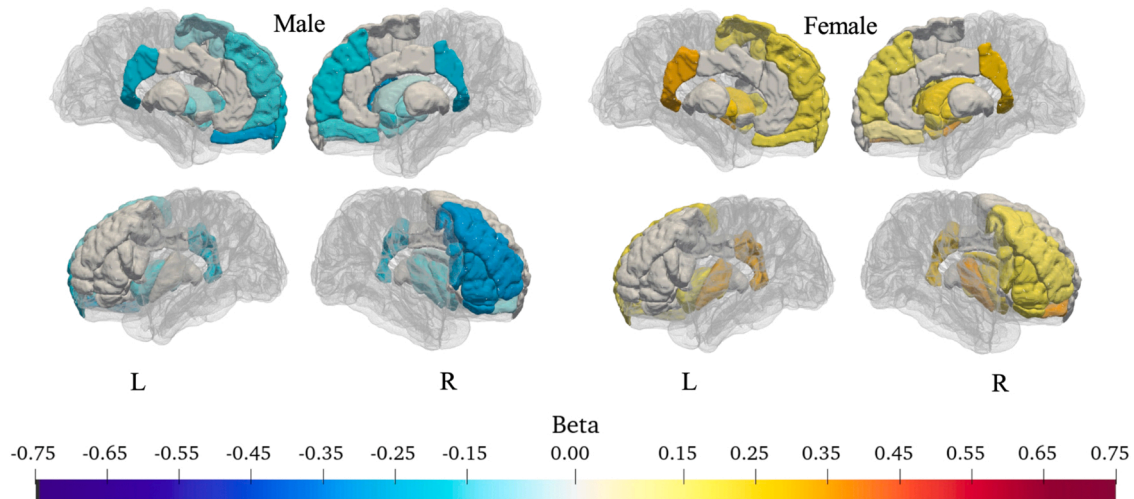


Fig. 4. Beta values for males (left) and females (right) in regions where models where significant 2-way interactions of sex and NEPSY-II Statue scores were the final model. For regions where a 2-way interaction was not the final model, the beta value was coded as 0.

and Advanced Test of Attention, which measures auditory attention, visual processing speed, visual-motor competence, and phonological awareness (Sung et al., 2021).

4.2. Differences by PAE status in male children

Unexposed males showed negative volume-executive function associations such that smaller volumes were associated with better executive function, while males with PAE showed positive volume-executive function associations across regions. The pattern in males with PAE is similar to the pattern previously observed in adults (Lean et al., 2017; Müller et al., 2015; Newman et al., 2007; Smolker et al., 2015; Takeuchi et al., 2013, 2017). The observation that male children with PAE show a more adult-like brain to executive function relationship than unexposed males potentially indicates accelerated brain maturation in males with PAE. Evidence for accelerated neural and behavioural maturation has been noted in response to early caregiving adversity (Callaghan and Tottenham, 2016) and has further been suggested as a potential outcome associated with adverse prenatal experiences such as maternal depression (Kling et al., 2023; Manning et al., 2022; Mareckova et al., 2020). However, brain maturation is likely affected by co-occurring pre- and postnatal exposures, which may have different and/or compounding effects on brain structure and function (Andre et al., 2020; Hassan et al., 2021). Moreover, our conclusion that males with PAE show an adult-like gray matter volume-executive function relationship is based on studies of adults *without* PAE. To our knowledge, studies of gray matter volume-executive function associations in adults with PAE have not yet been published. As such, our speculation that the pattern observed in males with PAE reflects a more mature pattern should be taken cautiously until further supportive evidence emerges.

4.3. Sex differences independent of PAE status

The dominant pattern we observed for the Statue measure was a two-way interaction of sex and Statue score, with males displaying negative and females displaying positive associations across regions, without exposure group differences. These relationships are consistent with the three-way interactions in the unexposed group seen for GEC, such that for males, smaller volumes were related to better executive function and for females, larger volumes were related to better executive function. We observed a conceptually similar relationship in three regions for the GEC: males had positive associations between volumes and GEC scores while females had negative associations between volumes and GEC scores, though it should be noted that our ability to interpret these models is limited because GEC *T* scores are normed by sex so, it is unclear whether this sex by GEC interaction is meaningful without further investigation.

While most previous studies did not specifically investigate sex differences, several noted no sex differences in volume-executive function relationships in adults (Takeuchi et al., 2013, 2017), and age-dependent sex differences in late childhood through adolescence (Sung et al., 2021). The latter study suggested that females show adult-like volume-executive function relationships at a younger age than males (Sung et al., 2021). In line with this, our finding from the Statue 2-way interaction analysis indicate that females show an association more similar to adults (*i.e.*, larger volume associated with better executive function) than males. This may reflect developmental effects, whereby the gray matter volume to performance-based executive function association matures faster in females than in males. The sex difference in the brain-behaviour relationship we observed in this study may also help to explain sex differences in performance in particular domains of executive function (Brocki and Bohlin, 2004; Grissom and Reyes, 2019; Wierenga et al., 2019).

While the patterns that we observed across the BRIEF/BRIEF-P GEC and Statue subtest were internally consistent, the number of significant effects and the regions where we observed them varied. The most

prevalent pattern for the Statue subtest was a sex effect, independent of PAE status (22 of 36 regions), while for the BRIEF/BRIEF-P GEC, 3-way interactions of PAE, sex and GEC were found. A potential explanation for this is that Statue is a child performance measure of motor persistence and inhibition, which is assessed in a structured setting while the GEC is a parent-reported measure of child behaviours in everyday settings. Parent reports and child performance-based measures of executive function assess different constructs of executive function (Ten Eycke and Dewey, 2016; Toplak et al., 2013). Performance-based measures of executive function may lack of ecological validity as they are conducted in environments with low distraction, high support, and structure which does not reflect the real-world application of executive function skills. Executive function skills are thought to be better assessed using parent and teacher rating scales (McAuley et al., 2010). Thus, our findings suggest that gray matter volume-executive function relationships vary by both PAE status and sex on a parent report of executive functions that assesses daily living functions, whereas performance on a structured motor persistence and inhibition task varies mainly by sex. In our study, Statue scores were not significantly different between participants with and without PAE, which is consistent with previous studies of very young children with and without PAE (Coles et al., 2021; Louw et al., 2021). Furthermore, children with PAE often exhibit inconsistency in their abilities depending on context, support, method of evaluation, and difficulty or complexity of a given task (Kodituwakku, 2007). Given the highly structured, simple nature of the Statue test, it is quite plausible that the limits of executive control are not reached, even for children with severe executive function impairments that are apparent in other settings (*ie.* at home as measured by the BRIEF/BRIEF-P). Our study suggests that at least when Statue measures are similar between individuals with and without PAE, most associations with gray matter volume are also similar between the exposure groups. However, in the few regions where we did find significant sex by PAE by Statue score interactions on gray matter volume, the direction of the relationships were conceptually similar to associations we observed for the BRIEF/BRIEF-P which bolsters confidence in the validity of our findings. It should be noted that others have found relationships with brain function, even when task performance was similar between children with and without PAE. For example, fMRI studies in adolescents with and without PAE, which used a performance measure of executive function to elicit associated brain activity, saw a similar level of task performance between groups but differences in the associated BOLD response (Kodali et al., 2017; Li et al., 2008; Ware et al., 2015).

4.4. Limitations

There are several limitations specific to our study. First, our alcohol exposed and unexposed samples differed in several ways, including sociodemographic variables and pre- and post-natal exposures. Children with PAE are generally more likely to have concurrent pre- and postnatal adverse exposures (Astley, 2010; Flannigan et al., 2021). Such co-occurring exposure can influence brain development and the development of executive function (Andre et al., 2020; Callaghan and Tottenham, 2016; Gautam, Warner, et al., 2015; Tooley et al., 2021), though it has been reported that PAE accounts for more variance in brain structure than other adverse exposures (Hemingway et al., 2020). Nonetheless, future studies with larger samples should attempt to better control for confounding factors such as the timing and amount of PAE as well as co-occurrence of other adverse exposures. Additionally, because the lack of PAE in the unexposed group was determined by parent report, it's possible that some parents mis-reported their alcohol use during pregnancy and that some of the children in the unexposed group were actually exposed to alcohol. Second, we used executive function measures from one time point in early childhood. While an individual's executive function ability as compared to similarly aged peers remains relatively stable over time for unexposed children (Gustavson et al., 2018; Miller and Marcovitch, 2015), difficulties in certain executive

function domains may disproportionately increase with age for children with PAE (Rasmussen and Bisanz, 2009) indicating that future studies of executive function in children with PAE would benefit from incorporating multiple measurements over time. Overall, the field would benefit from studies that examine the concurrent development of executive function and brain structure both by sex and by exposure status.

5. Conclusions

In this study, we characterized relationships between regional gray matter volumes and executive function, specifically a performance-based measure of inhibition and a parent-report measure of global executive function, in a longitudinal neuroimaging sample of young children with and without PAE. Our findings suggest that males with PAE, females with PAE, unexposed males, and unexposed females each display unique associations between gray matter volumes and a parent-report measure of executive function. Additionally, we found that males and females, regardless of PAE status, show different gray matter volume-executive function associations with a child performance-based measure of inhibition and motor persistence during the early childhood period. Our findings invite further study of sex differences in the gray matter volume-executive function associations across development and investigations of the nature of the gray matter volume-executive function associations in individuals with PAE across the lifespan.

Ethics statement

The studies involving humans were approved by Conjoint Health Research Ethics Board at the University of Calgary. The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent for participation in this study was provided by the participants' legal guardians/next of kin.

Funding Sources

The author(s) declare financial support was received for the research, authorship, and/or publication of this article. Data collection was supported by the Canadian Institutes of Health Research (grant numbers IHD-134090, MOP-136797), the Kids' Brain Health Network, and a grant from the Alberta Children's Hospital Research Institute. Salary support was provided by the Canada Research Chairs Program (CL) and a scholarship from the Cumming School of Medicine at the University of Calgary (ML).

CRedit authorship contribution statement

Madison Long: Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Catherine Lebel:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization. **Nils D. Forkert:** Writing – review & editing, Methodology. **Preeti Kar:** Writing – review & editing, Data curation. **Gerald F. Giesbrecht:** Writing – review & editing, Project administration, Funding acquisition. **Bennett A. Landman:** Writing – review & editing, Software, Methodology. **Deborah Dewey:** Writing – review & editing, Project administration, Funding acquisition. **Christina Tortorelli:** Writing – review & editing, Funding acquisition. **W. Ben Gibbard:** Writing – review & editing, Funding acquisition. **Yuankai Huo:** Writing – review & editing, Software, Methodology. **Carly A. McMorris:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of Competing Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.dcn.2025.101608.

Data availability

Data will be made available on request.

References

- Anderson, P., 2002. Assessment and development of executive function (EF) during childhood. *Child Neuropsychol.* 8 (2), 71–82. <https://doi.org/10.1076/chin.8.2.71.8724>.
- Andre, Q.R., McMorris, C.A., Kar, P., Ritter, C., Gibbard, W.B., Tortorelli, C., Lebel, C., 2020. Different brain profiles in children with prenatal alcohol exposure with or without early adverse exposures. *Hum. Brain Mapp.* 41 (15), 4375–4385. <https://doi.org/10.1002/hbm.25130>.
- Archibald, S.L., Fennema-Notestine, C., Gamst, A., Riley, E.P., Mattson, S.N., Jernigan, T.L., 2001. Brain dysmorphology in individuals with severe prenatal alcohol exposure. *Dev. Med. Child Neurol.* 43 (3), 148–154. <https://doi.org/10.1111/j.1469-8749.2001.tb00179.x>.
- Asman, A.J., Landman, B.A., 2013. Non-local statistical label fusion for multi-Atlas segmentation. *Med. Image Anal.* 17 (2), 194–208. <https://doi.org/10.1016/j.media.2012.10.002>.
- Asman, A.J., Landman, B.A., 2012. Formulating Spatially Varying Performance in the Statistical Fusion Framework. *IEEE Transactions on Medical Imaging*, 31(6), 1326–1336. *IEEE Transactions on Medical Imaging*. <https://doi.org/10.1109/TMI.2012.2190992>.
- Astley, S.J. (2010). PROFILE OF THE FIRST 1,400 PATIENTS RECEIVING DIAGNOSTIC EVALUATIONS FOR FETAL ALCOHOL SPECTRUM DISORDER AT THE WASHINGTON STATE FETAL ALCOHOL SYNDROME DIAGNOSTIC & PREVENTION NETWORK. 17.
- Astley, S.J., Aylward, E.H., Olson, H.C., Kerns, K., Brooks, A., Coggins, T.E., Davies, J., Dorn, S., Gendler, B., Jirikowic, T., Kraegel, P., Maravilla, K., Richards, T., 2009. Magnetic resonance imaging outcomes from a comprehensive magnetic resonance study of children with fetal alcohol spectrum disorders. *Alcohol. Clin. Exp. Res.* 33 (10), 1671–1689. <https://doi.org/10.1111/j.1530-0277.2009.01004.x>.
- Barch, D.M., Donohue, M.R., Elsayed, N.M., Gilbert, K., Harms, M.P., Hennefeld, L., Herzberg, M., Kandala, S., Karcher, N.R., Jackson, J.J., Luking, K.R., Rappaport, B.I., Sanders, A., Taylor, R., Tillman, R., Vogel, A.C., Whalen, D., Luby, J.L., 2022. Early childhood socioeconomic status and cognitive and adaptive outcomes at the transition to adulthood: the mediating role of gray matter development across five scan waves. *Biol. Psychiatry Cognit. Neurosci. Neuroimaging* 7 (1), 34–44. <https://doi.org/10.1016/j.bpsc.2021.07.002>.
- Barkley, R.A., 2010. Evaluating executive functioning deficits in everyday life. *ADHD Rep.* 18 (6), 9–10. <https://doi.org/10.1521/adhd.2010.18.6.9>.
- Bates, D., Mächler, M., Bolker, B., Walker, S., 2015. Fitting linear Mixed-Effects models using lme4. *Article 1. J. Stat. Softw.* 67 (1). <https://doi.org/10.18637/jss.v067.i01>.
- Bazin, P.-L., Pham, D.L., 2008. Homeomorphic brain image segmentation with topological and statistical atlases. *Med. Image Anal.* 12 (5), 616–625. <https://doi.org/10.1016/j.media.2008.06.008>.
- Benjamini, Y., Hochberg, Y., 1995. Controlling the false discovery rate: a practical and powerful approach to multiple testing. *J. R. Stat. Soc. Ser. B (Methodol.)* 57 (1), 289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>.
- Broccia, M., Munch, A., Hansen, B.M., Sørensen, K.K., Larsen, T., Strandberg-Larsen, K., Gerds, T.A., Torp-Pedersen, C., Kesmodel, U.S., 2023. Heavy prenatal alcohol exposure and overall morbidities: a danish nationwide cohort study from 1996 to 2018. *Lancet Public Health* 8 (1), e36–e46. [https://doi.org/10.1016/S2468-2667\(22\)00289-4](https://doi.org/10.1016/S2468-2667(22)00289-4).
- Brocki, K.C., Bohlin, G., 2004. Executive functions in children aged 6 to 13: a dimensional and developmental study. *Dev. Neuropsychol.* 26 (2), 571–593. <https://doi.org/10.1207/s15326942dn2602.3>.
- Burden, M.J., Andrew, C., Saint-Amour, D., Meintjes, E.M., Molteno, C.D., Hoyme, H.E., Robinson, L.K., Khaole, N., Nelson, C.A., Jacobson, J.L., Jacobson, S.W., 2009. The effects of fetal alcohol syndrome on response execution and inhibition: an Event-Related potential study. *Alcohol. Clin. Exp. Res.* 33 (11), 1994–2004. <https://doi.org/10.1111/j.1530-0277.2009.01038.x>.
- Burden, M.J., Westerlund, A., Muckle, G., Dodge, N., Dewailly, E., Nelson, C.A., Jacobson, S.W., Jacobson, J.L., 2011. The effects of maternal binge drinking during pregnancy on neural correlates of response inhibition and memory in childhood. *Alcohol. Clin. Exp. Res.* 35 (1), 69–82. <https://doi.org/10.1111/j.1530-0277.2010.01323.x>.
- Calkins, S.D., Marcovitch, S., 2010. Emotion regulation and executive functioning in early development: integrated mechanisms of control supporting adaptive functioning. *Child Development at the Intersection of Emotion and Cognition*. American Psychological Association, pp. 37–57. <https://doi.org/10.1037/12059-003>.
- Callaghan, B.L., Tottenham, N., 2016. The stress acceleration hypothesis: effects of early-life adversity on emotion circuits and behavior. *Curr. Opin. Behav. Sci.* 7, 76–81. <https://doi.org/10.1016/j.cobeha.2015.11.018>.
- Clark, C., Prior, M., Kinsella, G., 2002. The relationship between executive function abilities, adaptive behaviour, and academic achievement in children with

- externalising behaviour problems. *J. Child Psychol. Psychiatry* 43 (6), 785–796. <https://doi.org/10.1111/1469-7610.00084>.
- Clark, S.V., Semmel, E.S., Aleksonis, H.A., Steinberg, S.N., King, T.Z., 2021. Cerebellar-Subcortical-cortical systems as modulators of cognitive functions. *Neuropsychol. Rev.* 31 (3), 422–446. <https://doi.org/10.1007/s11065-020-09465-1>.
- Coles, C.D., Kable, J.A., Granovska, I.V., Pashtepa, A.O., Wertschke, W., Chambers, C.D., The, C.I.F.A.S.D., 2021. Measurement of neurodevelopmental effects of prenatal alcohol exposure in Ukrainian preschool children. *Child Neuropsychol.* 27 (8), 1088–1103. <https://doi.org/10.1080/09297049.2021.1919298>.
- Cook, J.L., Green, C.R., Lilley, C.M., Anderson, S.M., Baldwin, M.E., Chudley, A.E., Conry, J.L., LeBlanc, N., Look, C.A., Lutke, J., Mallon, B.F., McFarlane, A.A., Temple, V.K., Rosales, T., 2016. Fetal alcohol spectrum disorder: a guideline for diagnosis across the lifespan. *CMAJ* 188 (3), 191–197. <https://doi.org/10.1503/cmaj.141593>.
- Cortese, B.M., Moore, G.J., Bailey, B.A., Jacobson, S.W., Delaney-Black, V., Hannigan, J.H., 2006. Magnetic resonance and spectroscopic imaging in prenatal alcohol-exposed children: preliminary findings in the caudate nucleus. *Neurotoxicol. Teratol.* 28 (5), 597–606. <https://doi.org/10.1016/j.ntt.2006.08.002>.
- Cristofori, I., Cohen-Zimmerman, S., Grafman, J., 2019. Chapter 11—executive functions. In: D'Esposito, M., Grafman, J.H. (Eds.), *Handbook of Clinical Neurology*. Elsevier, pp. 197–219. <https://doi.org/10.1016/B978-0-12-804281-6.00011-2>. Vol. 163.
- DeCasien, A.R., Guma, E., Liu, S., Raznahan, A., 2022. Sex differences in the human brain: a roadmap for more careful analysis and interpretation of a biological reality. *Biol. Sex. Differ.* 13 (1), 43. <https://doi.org/10.1186/s13293-022-00448-w>.
- Diamond, A., 2013. Executive functions. *Annu. Rev. Psychol.* 64, 135–168. <https://doi.org/10.1146/annurev-psych-113011-143750>.
- Donald, K.A., Eastman, E., Howells, F.M., Adnams, C., Riley, E.P., Woods, R.P., Narr, K.L., Stein, D.J., 2015. Neuroimaging effects of prenatal alcohol exposure on the developing human brain: A magnetic resonance imaging review. *Acta Neuropsychiatr.* 27 (5), 251–269. <https://doi.org/10.1017/neu.2015.12>.
- Fiske, A., Holmboe, K., 2019. Neural substrates of early executive function development. *Dev. Rev.* 52, 42–62. <https://doi.org/10.1016/j.dr.2019.100866>.
- Flannigan, K., Kapsi, A., Pei, J., Murdoch, I., Andrew, G., Rasmussen, C., 2021. Characterizing adverse childhood experiences among children and adolescents with prenatal alcohol exposure and fetal alcohol spectrum disorder. *Child Abus. Negl.* 112, 104888. <https://doi.org/10.1016/j.chiabu.2020.104888>.
- Flannigan, K., Poole, N., Cook, J., Unsworth, K., 2023. Sex-related differences among individuals assessed for fetal alcohol spectrum disorder in Canada. *Alcohol. Clin. Exp. Res.* 47 (3), 613–623. <https://doi.org/10.1111/acer.15017>.
- Friedman, N.P., Miyake, A., 2017. Unity and diversity of executive functions: individual differences as a window on cognitive structure. *Cortex* 86, 186–204. <https://doi.org/10.1016/j.cortex.2016.04.023>.
- Friedman, N.P., Robbins, T.W., 2022. The role of prefrontal cortex in cognitive control and executive function. *Neuropsychopharmacology* 47 (1), 72–89. <https://doi.org/10.1038/s41386-021-01132-0>.
- Gaillard, A., Fehring, D.J., Rossell, S.L., 2021a. A systematic review and meta-analysis of behavioural sex differences in executive control. *Eur. J. Neurosci.* 53 (2), 519–542. <https://doi.org/10.1111/ejn.14946>.
- Gaillard, A., Fehring, D.J., Rossell, S.L., 2021b. Sex differences in executive control: a systematic review of functional neuroimaging studies. *Eur. J. Neurosci.* 53 (8), 2592–2611. <https://doi.org/10.1111/ejn.15107>.
- Gautam, P., Lebel, C., Narr, K.L., Mattson, S.N., May, P.A., Adnams, C.M., Riley, E.P., Jones, K.L., Kan, E.C., Sowell, E.R., 2015. Volume changes and brain-behavior relationships in White matter and subcortical gray matter in children with prenatal alcohol exposure: volume changes and Brain-Behavior in children with PAE. *Hum. Brain Mapp.* 36 (6), 2318–2329. <https://doi.org/10.1002/hbm.22772>.
- Gautam, P., Nuñez, S.C., Narr, K.L., Kan, E.C., Sowell, E.R., 2014. Effects of prenatal alcohol exposure on the development of White matter volume and change in executive function. *NeuroImage Clin.* 5, 19–27. <https://doi.org/10.1016/j.nicl.2014.05.010>.
- Gautam, P., Warner, T.D., Kan, E.C., Sowell, E.R., 2015. Executive function and cortical thickness in youths prenatally exposed to cocaine, alcohol and tobacco. *Dev. Cogn. Neurosci.* 1, 155–165.
- Gioia, G.A., Espy, K.A., Isquith, P.K., 2003. BRIEF-P: behavior rating inventory of executive function—preschool version. *Psychol. Assess. Resour. (PAR)*.
- Gioia, G.A., Isquith, P.K., Guy, S.C., Kenworthy, L., 2000. Test review behavior rating inventory of executive function. *Child Neuropsychol.* 6 (3), 235–238.
- Gioia, G.A., Isquith, P.K., Retzlaff, P.D., Espy, K.A., 2002. Confirmatory factor analysis of the behavior rating inventory of executive function (BRIEF) in a clinical sample. *Child Neuropsychol.* 8 (4), 249–257. <https://doi.org/10.1076/chin.8.4.249.13513>.
- Grissom, N.M., Reyes, T.M., 2019. Let's call the whole thing off: evaluating gender and sex differences in executive function. Article 1. *Neuropsychopharmacology* 44 (1). <https://doi.org/10.1038/s41386-018-0179-5>.
- Gustavson, D.E., Panizzon, M.S., Elman, J.A., Franz, C.E., Reynolds, C.A., Jacobson, K.C., Friedman, N.P., Xian, H., Toomey, R., Lyons, M.J., Kremen, W.S., 2018. Stability of genetic and environmental influences on executive functions in midlife. *Psychol. Aging* 33 (2), 219–231. <https://doi.org/10.1037/pag0000230>.
- Han, X., Xu, C., Tosun, D., Prince, J.L., 2001. Cortical surface reconstruction using a topology preserving geometric deformable model. *Proc. IEEE Workshop Math. Methods Biomed. Image Anal. (MMBIA 2001)* 213–220. <https://doi.org/10.1109/MMBIA.2001.991736>.
- Han, X., & Fischl, B., 2007. Atlas Renormalization for Improved Brain MR Image Segmentation Across Scanner Platforms. *IEEE Transactions on Medical Imaging*, 26 (4), 479–486. <https://doi.org/10.1109/TMI.2007.893282>.
- Han, X., Xu, C., Braga-Neto, U., & Prince, J.L., 2002. Topology correction in brain cortex segmentation using a multiscale, graph-based algorithm. *IEEE Transactions on Medical Imaging*, 21(2), 109–121. <https://doi.org/10.1109/42.993130>.
- Hassan, R., MacMillan, H.L., Saigal, S., Schmidt, L.A., 2021. Brain, interrupted: alpha/delta EEG ratio in survivors of pre- and post-natal adversity. *Int. J. Neurosci.* 132 (1), 31–37. <https://doi.org/10.1080/00207454.2020.1797724>.
- Hemingway, S.J.A., Davies, J.K., Jirikovic, T., Olson, E.M., 2020. What proportion of the brain structural and functional abnormalities observed among children with fetal alcohol spectrum disorder is explained by their prenatal alcohol exposure and their other prenatal and postnatal risks? *Adv. Pediatr. Res.* 16.
- Huo, Y., Bao, S., Parvathaneni, P., & Landman, B.A. (2017). Improved Stability of Whole Brain Surface Parcellation with Multi-Atlas Segmentation. *arXiv:1712.00543 [Cs]*. <http://arxiv.org/abs/1712.00543>.
- Huo, Y., Carass, A., Resnick, S.M., Pham, D.L., Prince, J.L., & Landman, B.A. (2016). *Combining multi-atlas segmentation with brain surface estimation* (M. A. Styner & E. D. Angelini, Eds.; p. 97840E). <https://doi.org/10.1117/12.2216604>.
- Huo, Y., Plassard, A.J., Carass, A., Resnick, S.M., Pham, D.L., Prince, J.L., Landman, B.A., 2016. Consistent cortical reconstruction and methods: APron cohort study. *NeuroImage* 138, 197–210. <https://doi.org/10.1016/j.neuroimage.2016.05.030>.
- Kaplan, B.J., Giesbrecht, G.F., Leung, B.M.Y., Field, C.J., Dewey, D., Bell, R.C., Manca, D. P., O'Beirne, M., Johnston, D.W., Pop, V.J., Singhal, N., Gagnon, L., Bernier, F.P., Eliasziw, M., McCargar, L.J., Kooistra, L., Farmer, A., Cantell, M., Goonewardene, L., APron Study Team, the, 2014. The alberta pregnancy outcomes and nutrition (APron) cohort study: rationale and methods: APron cohort study. *Matern. Child Nutr.* 10 (1), 44–60. <https://doi.org/10.1111/j.1740-8709.2012.00433.x>.
- Khouri, J.E., Milligan, K., Girard, T.A., 2015. Executive Functioning in Children and Adolescents Prenatally Exposed to Alcohol: A Meta-Analytic Review. *Neuropsychol. Rev.* 25 (2), 149–170. <https://doi.org/10.1007/s11065-015-9289-6>.
- Klein, A., Dal Canton, T., Ghosh, S.S., Landman, B., Lee, J., & Worth, A. (2010). Open labels: Online feedback for a public resource of manually labeled brain images. 16th Annual Meeting for the Organization of Human Brain Mapping.
- Kling, J.L., Mistry-Patel, S., Peoples, S.G., Caldera, D.R., Brooker, R.J., 2023. Prenatal maternal depression predicts neural maturation and negative emotion in infants. *Infant Behav. Dev.* 70, 101802. <https://doi.org/10.1016/j.infbeh.2022.101802>.
- Kodali, V.N., Jacobson, J.L., Lindinger, N.M., Dodge, N.C., Molteno, C.D., Meintjes, E.M., Jacobson, S.W., 2017. Differential recruitment of brain regions during response inhibition in children prenatally exposed to alcohol. *Alcohol. Clin. Exp. Res.* 41 (2), 334–344. <https://doi.org/10.1111/acer.13307>.
- Koditwakkul, P.W., 2007. Defining the behavioral phenotype in children with fetal alcohol spectrum disorders: a review. *Neurosci. Biobehav. Rev.* 31 (2), 192–201. <https://doi.org/10.1016/j.neubiorev.2006.06.020>.
- Koditwakkul, P.W., Kalberg, W., May, P.A., 2001. The Effects of Prenatal Alcohol Exposure on Executive Functioning. *Alcohol Res. Health* 25 (3), 192–198.
- Korkman, M., Kirk, U., & Kemp, S. (2007). *NEPSY II: Clinical and interpretive manual*. Harcourt Assessment, PsychCorp. <https://researchportal.helsinki.fi/en/publications/nepsy-ii-clinical-and-interpretive-manual>.
- Lean, R.E., Melzer, T.R., Bora, S., Watts, R., Woodward, L.J., 2017. Attention and regional gray matter development in very preterm children at age 12 years. *J. Int. Neuropsychol.* 23 (7), 539–550. <https://doi.org/10.1017/S155617717000388>.
- Lebel, C., Roussotte, F., Sowell, E.R., 2011. Imaging the Impact of Prenatal Alcohol Exposure on the Structure of the Developing Human Brain. *Neuropsychol. Rev.* 21 (2), 102–118. <https://doi.org/10.1007/s11065-011-9163-0>.
- Lenroot, R.K., Gogtay, N., Greenstein, D.K., Wells, E.M., Wallace, G.L., Clasen, L.S., Blumenthal, J.D., Lerch, J., Zijdenbos, A.P., Evans, A.C., Thompson, P.M., Giedd, J. N., 2007. Sexual dimorphism of brain developmental trajectories during childhood and adolescence. *NeuroImage* 36 (4), 1065–1073. <https://doi.org/10.1016/j.neuroimage.2007.03.053>.
- Lenth, R.V. (2017). *emmeans: Estimated Marginal Means, aka Least-Squares Means* (p. 1.10.2) [Dataset]. <https://doi.org/10.32614/CRAN.package.emmeans>.
- Li, Z., Coles, C.D., Lynch, M.E., Ma, X., Peltier, S., Hu, X., 2008. Occipital-temporal reduction and sustained visual attention deficit in prenatal alcohol exposed adults. *Brain Imaging Behav.* 2 (1), 39–48. <https://doi.org/10.1007/s11682-007-9013-0>.
- Long, M., Kar, P., Forkert, N.D., Landman, B.A., Gibbard, W.B., Tortorelli, C., McMorris, C.A., Huo, Y., Lebel, C.A., 2024. Sex and age effects on gray matter volume trajectories in young children with prenatal alcohol exposure. *Front. Hum. Neurosci.* 18. <https://doi.org/10.3389/fnhum.2024.1379959>.
- Long, M., Ostertag, C., Reynolds, J.E., Zheng, J., Landman, B., Huo, Y., Forkert, N.D., Lebel, C., 2024. Few sex differences in regional gray matter volume growth trajectories across early childhood. *Imaging Neurosci.* 2, 1–26. https://doi.org/10.1162/imag_a.00154.
- Louw, J.G., Heerden, A. van, Olivier, L., Lambrechts, T., Broodryk, M., Bunge, L., Vosloo, M., Tomlinson, M., 2021. Executive function after prenatal alcohol exposure in children in a South African population: cross-sectional study. *JMIR Form. Res.* 5 (7), e20658. <https://doi.org/10.2196/20658>.
- Manning, K.Y., Long, X., Watts, D., Tomfohr-Madsen, L., Giesbrecht, G.F., Lebel, C., 2022. Prenatal maternal distress during the COVID-19 pandemic and associations with infant brain connectivity. *Biol. Psychiatry* 92 (9), 701–708. <https://doi.org/10.1016/j.biopsych.2022.05.011>.
- Mareckova, K., Marecek, R., Andryskova, L., Brazdil, M., Nikolova, Y.S., 2020. Maternal depressive symptoms during pregnancy and brain age in young adult offspring: findings from a prenatal birth cohort. *Cereb. Cortex* 30 (7), 3991–3999. <https://doi.org/10.1093/cercor/bhaa014>.

- Mattson, S.N., Bernes, G.A., Doyle, L.R., 2019. Fetal Alcohol Spectrum Disorders: A Review of the Neurobehavioral Deficits Associated With Prenatal Alcohol Exposure. *Alcoholol.: Clin. Exp. Res.* 43 (6), 1046–1062. <https://doi.org/10.1111/acer.14040>.
- Mattson, S.N., Riley, E.P., Jernigan, T.L., Ehlers, C.L., Delis, D.C., Jones, K.L., Stern, C., Johnson, K.A., Hesselink, J.R., Bellugi, U., 1992. Fetal alcohol syndrome: a case report of neuropsychological, MRI and EEG assessment of two children. *Alcoholol. Clin. Exp. Res.* 16 (5), 1001–1003. <https://doi.org/10.1111/j.1530-0277.1992.tb01909.x>.
- Mattson, S.N., Riley, E.P., 1998. A Review of the Neurobehavioral Deficits in Children with Fetal Alcohol Syndrome or Prenatal Exposure to Alcohol. *Alcoholol.: Clin. Exp. Res.* 22 (2), 279–294. <https://doi.org/10.1111/j.1530-0277.1998.tb03651.x>.
- Mattson, S.N., Riley, E.P., Sowell, E.R., Jernigan, T.L., Sobel, D.F., Jones, K.L., 1996. A decrease in the size of the basal ganglia in children with fetal alcohol syndrome. *Alcoholol. Clin. Exp. Res.* 20 (6), 1088–1093. <https://doi.org/10.1111/j.1530-0277.1996.tb01951.x>.
- May, P.A., Chambers, C.D., Kalberg, W.O., Zellner, J., Feldman, H., Buckley, D., Kopald, D., Hasken, J.M., Xu, R., Honerkamp-Smith, G., Taras, H., Manning, M.A., Robinson, L.K., Adam, M.P., Abdul-Rahman, O., Vaux, K., Jewett, T., Elliott, A.J., Kable, J.A., Hoyme, H.E., 2018. Prevalence of fetal alcohol spectrum disorders in 4 US communities. *JAMA* 319 (5), 474–482. <https://doi.org/10.1001/jama.2017.21896>.
- McAlonan, G.M., Cheung, V., Chua, S.E., Oosterlaan, J., Hung, S., Tang, C., Lee, C., Kwong, S., Ho, T., Cheung, C., Suckling, J., Leung, P.W.L., 2009. Age-related grey matter volume correlates of response inhibition and shifting in attention-deficit hyperactivity disorder. *Br. J. Psychiatry* 194 (2), 123–129. <https://doi.org/10.1192/bjp.bp.108.051359>.
- McAuley, T., Chen, S., Goos, L., Schachar, R., Crosbie, J., 2010. Is the BRIEF more strongly associated with measures of impairment or executive function? *J. Int. Neuropsychol. Soc. JINS* 16, 495–505. <https://doi.org/10.1017/S1355617710000093>.
- Menon, V., D'Esposito, M., 2022. The role of PFC networks in cognitive control and executive function. *Neuropsychopharmacology* 47 (1), 90–103. <https://doi.org/10.1038/s41386-021-01152-w>.
- Migliorini, R., Moore, E.M., Glass, L., Infante, M.A., Tapert, S.F., Jones, K.L., Mattson, S. N., Riley, E.P., 2015. Anterior cingulate cortex surface area relates to behavioral inhibition in adolescents with and without heavy prenatal alcohol exposure. *Behav. Brain Res.* 292, 26–35. <https://doi.org/10.1016/j.bbr.2015.05.037>.
- Miller, S.E., Marcovitch, S., 2015. Examining executive function in the second year of life: coherence, stability, and relations to joint attention and language. *Dev. Psychol.* 51 (1), 101–114. <https://doi.org/10.1037/a0038359>.
- Müller, V.I., Langner, R., Cieslik, E.C., Rottschy, C., Eickhoff, S.B., 2015. Interindividual differences in cognitive flexibility: influence of gray matter volume, functional connectivity and trait impulsivity. *Brain Struct. Funct.* 220 (4), 2401–2414. <https://doi.org/10.1007/s00429-014-0797-6>.
- Nardelli, A., Lebel, C., Rasmussen, C., Andrew, G., Beaulieu, C., 2011. Extensive deep gray matter volume reductions in children and adolescents with fetal alcohol spectrum disorders. *Alcoholol. Clin. Exp. Res.* 35 (8), 1404–1417. <https://doi.org/10.1111/j.1530-0277.2011.01476.x>.
- Newman, L.M., Trivedi, M.A., Bendlin, B.B., Ries, M.L., Johnson, S.C., 2007. The relationship between gray matter morphometry and neuropsychological performance in a large sample of cognitively healthy adults. *Brain Imaging Behav.* 1 (1), 3–10. <https://doi.org/10.1007/s11682-007-9000-5>.
- NiftyReg—C.M.I.C. (n.d.). Retrieved December 15, 2023, from <http://cmictg.cs.ucl.ac.uk/wiki/index.php/NiftyReg>.
- Popova, S., Charness, M.E., Burd, L., Crawford, A., Hoyme, H.E., Mukherjee, R.A.S., Riley, E.P., Elliott, E.J., 2023. Fetal alcohol spectrum disorders. Article 1. *Nat. Rev. Dis. Prim.* 9 (1). <https://doi.org/10.1038/s41572-023-00420-x>.
- Rajaprakash, M., Chakravarty, M.M., Lerch, J.P., Rovet, J., 2014. Cortical morphology in children with alcohol-related neurodevelopmental disorder. *Brain Behav.* 4 (1), 41–50. <https://doi.org/10.1002/brb3.191>.
- Rasmussen, C., Bisanz, J., 2009. Executive functioning in children with fetal alcohol spectrum disorders: profiles and Age-Related differences. *Child Neuropsychol.* 15 (3), 201–215. <https://doi.org/10.1080/09297040802385400>.
- Rasmussen, C., Tamana, S., Baugh, L., Andrew, G., Tough, S., Zwaigenbaum, L., 2013. Neuropsychological impairments on the NEPSY-II among children with FASD. *Child Neuropsychol.* 19 (4), 337–349. <https://doi.org/10.1080/09297049.2012.658768>.
- Rasmussen, C., McAuley, R., Andrew, G., 2007. PARENTAL RATINGS OF CHILDREN WITH FETAL ALCOHOL SPECTRUM DISORDER ON THE BEHAVIOR RATING INVENTORY OF EXECUTIVE FUNCTION (BRIEF), 8.
- Reynolds, J.E., Long, X., Paniukov, D., Bagshawe, M., Lebel, C., 2020. Calgary preschool magnetic resonance imaging (MRI) dataset. *Data Brief.* 29, 105224. <https://doi.org/10.1016/j.dib.2020.105224>.
- Rockhold, M.N., Krueger, A.M., de Water, E., Lindgren, C.W., Sandness, K.E., Eckerle, J. K., Schumacher, M.J., Fink, B.A., Boys, C.J., Carlson, S.M., Fuglestad, A.J., Mattson, S.N., Jones, K.L., Riley, E.P., Wojniak, J.R., 2021. Executive and Social Functioning Across Development in Children and Adolescents With Prenatal Alcohol Exposure. *Alcoholol.: Clin. Exp. Res.* 45 (2), 457–469. <https://doi.org/10.1111/acer.14538>.
- Roussotte, F.F., Sulik, K.K., Mattson, S.N., Riley, E.P., Jones, K.L., Adnams, C.M., May, P. A., O'Connor, M.J., Narr, K.L., Sowell, E.R., 2011. Regional brain volume reductions relate to facial dysmorphology and neurocognitive function in fetal alcohol spectrum disorders. *Hum. Brain Mapp.* 33 (4), 920–937. <https://doi.org/10.1002/hbm.21260>.
- Sethian, J.A. (1999). Level Set Methods and Fast Marching Methods: Evolving Interfaces in Computational Geometry, Fluid Mechanics, Computer Vision, and Materials Science. Cambridge University Press.
- Sherman, E., Brooks, B., 2010. Behavior rating inventory of executive function – preschool version (BRIEF-P): test review and clinical guidelines for use. *Child Neuropsychology: A Journal Normal Abnormal Development Childhood Adolescence* 16, 1–17. <https://doi.org/10.1080/09297041003679344>.
- Smolker, H.R., Depue, B.E., Reineberg, A.E., Orr, J.M., Banich, M.T., 2015. Individual differences in regional prefrontal gray matter morphometry and fractional anisotropy are associated with different constructs of executive function. *Brain Struct. Funct.* 220 (3), 1291–1306. <https://doi.org/10.1007/s00429-014-0723-y>.
- Subramoney, S., Joshi, S.H., Wedderburn, C.J., Lee, D., Roos, A., Woods, R.P., Zar, H.J., Narr, K.L., Stein, D.J., Donald, K.A., 2022. The impact of prenatal alcohol exposure on gray matter volume and cortical surface area of 2 to 3-year-old children in a South African birth cohort. *Alcoholol. Clin. Exp. Res.* 46 (7), 1233–1247. <https://doi.org/10.1111/acer.14873>.
- Sung, D., Park, B., Kim, B., Kim, H., Jung, K.-I., Lee, S.-Y., Kim, B.-N., Park, S., Park, M.-H., 2021. Gray matter volume in the developing frontal lobe and its relationship with executive function in late childhood and adolescence: a Community-Based study. *Front. Psychiatry* 12. <https://doi.org/10.3389/fpsy.2021.686174>.
- Takeuchi, H., Taki, Y., Nouchi, R., Yokoyama, R., Kotozaki, Y., Nakagawa, S., Sekiguchi, A., Iizuka, K., Yamamoto, Y., Hanawa, S., Araki, T., Miyauchi, C.M., Shinada, T., Sakaki, K., Sassa, Y., Nozawa, T., Ikeda, S., Yokota, S., Daniele, M., Kawashima, R., 2017. Global associations between regional gray matter volume and diverse complex cognitive functions: evidence from a large sample study. *Sci. Rep.* 7 (1), 10014. <https://doi.org/10.1038/s41598-017-10104-8>.
- Takeuchi, H., Taki, Y., Sassa, Y., Hashizume, H., Sekiguchi, A., Fukushima, A., Kawashima, R., 2013. Brain structures associated with executive functions during everyday events in a non-clinical sample. *Brain Struct. Funct.* 218 (4), 1017–1032. <https://doi.org/10.1007/s00429-012-0444-z>.
- Ten Eyck, K.D., Dewey, D., 2016. Parent-report and performance-based measures of executive function assess different constructs. *Child Neuropsychol.* 22 (8), 889–906. <https://doi.org/10.1080/09297049.2015.1065961>.
- Terasaki, L.S., Gomez, J., Schwarz, J.M., 2016. An examination of sex differences in the effects of early-life opiate and alcohol exposure. *Philos. Trans. R. Soc. B: Biol. Sci.* 371 (1688), 20150123. <https://doi.org/10.1098/rstb.2015.0123>.
- Thieba, C., Frayne, A., Walton, M., Mah, A., Benischek, A., Dewey, D., Lebel, C., 2018. Factors associated with successful MRI scanning in unsedated young children. *Front. Pediatr.* 6. <https://doi.org/10.3389/fped.2018.00146>.
- Tooley, U.A., Bassett, D.S., Mackey, A.P., 2021. Environmental influences on the pace of brain development. *Nat. Rev. Neurosci.* 22 (6), 372–384. <https://doi.org/10.1038/s41583-021-00457-5>.
- Toplak, M.E., West, R.F., Stanovich, K.E., 2013. Practitioner review: do performance-based measures and ratings of executive function assess the same construct? *J. Child Psychol. Psychiatry* 54 (2), 131–143. <https://doi.org/10.1111/jcpp.12001>.
- Treit, S., Chen, Z., Zhou, D., Baugh, L., Rasmussen, C., Andrew, G., Pei, J., Beaulieu, C., 2017. Sexual dimorphism of volume reduction but not cognitive deficit in fetal alcohol spectrum disorders: a combined diffusion tensor imaging, cortical thickness and brain volume study. *NeuroImage Clin.* 15, 284–297. <https://doi.org/10.1016/j.nicl.2017.05.006>.
- Treit, S., Lebel, C., Baugh, L., Rasmussen, C., Andrew, G., Beaulieu, C., 2013. Longitudinal MRI reveals altered trajectory of brain development during childhood and adolescence in fetal alcohol spectrum disorders. *Journal Neuroscience Official Journal Society Neuroscience* 33 (24), 10098–10109. <https://doi.org/10.1523/JNEUROSCI.5004-12.2013>.
- Tustison, N.J., Avants, B.B., Cook, P.A., Zheng, Y., Egan, A., Yushkevich, P.A., Gee, J.C., 2010. N4ITK: improved N3 bias correction. *IEEE Trans. Med. Imaging* 29 (6), 1310–1320. <https://doi.org/10.1109/TMI.2010.2046908>.
- Ware, A.L., Crocker, N., O'Brien, J.W., Dewese, B.N., Roesch, S.C., Coles, C.D., Kable, J. A., May, P.A., Kalberg, W.O., Sowell, E.R., Jones, K.L., Riley, E.P., Mattson, S.N., 2012. Executive function predicts adaptive behavior in children with histories of heavy prenatal alcohol exposure and attention-deficit/hyperactivity disorder. *Alcoholol. Clin. Exp. Res.* 36 (8), 1431–1441. <https://doi.org/10.1111/j.1530-0277.2011.01718.x>.
- Ware, A.L., Infante, M.A., O'Brien, J.W., Tapert, S.F., Jones, K.L., Riley, E.P., Mattson, S. N., 2015. An fMRI study of behavioral response inhibition in adolescents with and without histories of heavy prenatal alcohol exposure. *Behav. Brain Res.* 1, 137–146.
- Weinberg, J., Sliwowska, J.H., Lan, N., Hellemans, K.G.C., 2008. Prenatal Alcohol Exposure: Foetal Programming, the Hypothalamic-Pituitary-Adrenal Axis and Sex Differences in Outcome. *J. Neuroendocrinol.* 20 (4), 470–488. <https://doi.org/10.1111/j.1365-2826.2008.01669.x>.
- Wierenga, L.M., Bos, M.G.N., van Rossum, F., Crone, E.A., 2019. Sex effects on development of brain structure and executive functions: greater variance than mean effects. *J. Cogn. Neurosci.* 31 (5), 730–753. https://doi.org/10.1162/jocn_a.01375.
- Wierenga, L.M., Langen, M., Oranje, B., Durston, S., 2014. Unique developmental trajectories of cortical thickness and surface area. *NeuroImage* 87, 120–126. <https://doi.org/10.1016/j.neuroimage.2013.11.010>.
- Yuan, P., Raz, N., 2014. Prefrontal cortex and executive functions in healthy adults: a meta-analysis of structural neuroimaging studies. *Neurosci. Biobehav. Rev.* 42, 180–192. <https://doi.org/10.1016/j.neubiorev.2014.02.005>.
- Zelazo, P.D., & Müller, U. (2010). Executive Function in Typical and Atypical Development. 574–603. <https://doi.org/10.1002/9781444325485.ch22>.
- Zelazo, P.D., Müller, U., Frye, D., Marcovitch, S., Argitis, G., Boseovski, J., Chiang, J.K., Hongwanishkul, D., Schuster, B.V., Sutherland, A., Carlson, S.M., 2003. The development of executive function in early childhood. *Monogr. Soc. Res. Child Dev.* 68 (3), i-151.